

Ethical Hacking Task 01: Information Gathering & Reconnaissance

Part A: Target Selection

- **Website Name:** kristujayanti.edu
- **URL:** <https://www.kristujayanti.edu.in/>
- **Reason for choosing the target:** College Website.

Part B: WHOIS Lookup

You will need to use a public WHOIS lookup to find the following details for your target domain:

Domain Name: kristujayanti.edu.in

Registrar: ERNET India

Registration Date: 2012-04-20

Expiry Date: 2035-04-20

Name Servers: ns-1333.awsdns-38.org
ns-265.awsdns-33.com
ns-812.awsdns-37.net
ns-1857.awsdns-40.co.uk

Domain Status: OK

Part C: DNS Enumeration

Using a tool like nslookup, dig, or an online DNS checker (like MXToolbox), query your domain to identify the following records:

- **A Record:**

A records

IPv4 address	Revalidate in
> 13.235.5.4 For CNAME kristujayanti.edu.in.	1m
> 13.232.185.200 For CNAME kristujayanti.edu.in.	1m

MX Record:

MX records

Mail server	Priority	Revalidate in
inbound-smtp.us-east-1.amazonaws.com. For CNAME kristujayanti.edu.in.	10 Primary	1m

NS Record:
NS records

Name server	Revalidate in
ns-1333.awsdns-38.org. For CNAME kristujayanti.edu.in.	1h
ns-1857.awsdns-40.co.uk. For CNAME kristujayanti.edu.in.	1h
ns-265.awsdns-33.com. For CNAME kristujayanti.edu.in. 	1h
ns-812.awsdns-37.net. For CNAME kristujayanti.edu.in.	1h

- TXT Record:**
TXT records

Site ownership verification



Google

VOrsWwWwEhAGFZkrn
d4U1jeqPgWzIFyoBiXUF
BGL3ig

Revalidate in 1m



Google

k0WiuRMXif5j3oqdug2
npE7j5RF7yz7bUarZgEb
Ldil

Revalidate in 1m

Part D: Website Technology Identification

Summary of Technologies Identified:

The reconnaissance indicates that the application is a modern, single-page or server-side rendered application. The primary programming language utilized is JavaScript/TypeScript, heavily relying on the React.js library and the Next.js framework for UI rendering and routing. Styling is managed via the Tailwind CSS framework.

Part E: HTTP Security Headers

Header	Present (Yes/No)	Purpose
Content-Security-Policy (CSP)	Yes	Prevents Cross-Site Scripting (XSS) and data injection attacks by specifying which dynamic resources are allowed to load.

Header	Present (Yes/No)	Purpose
X-Frame-Options	Yes	Protects against Clickjacking by controlling whether the site can be rendered inside a <frame>, <iframe>, or <object>.
X-Content-Type-Options	Yes	Stops the browser from trying to MIME-sniff the content type, enforcing the declared content type to prevent certain types of execution attacks.
Strict-Transport-Security (HSTS)	Yes	Forces the browser to communicate with the server exclusively over secure HTTPS connections, mitigating downgrade attacks.
Referrer-Policy	Yes	Controls how much referrer information (the URL of the previous page) is included with requests made from your site.

Part F: Robots.txt & Sitemap Analysis

- **Does the website have a robots.txt file?**

YES

Does it have a sitemap?

YES

- **What information can be learned from these files?**

The robots.txt file reveals which directories or paths the site owner wants to hide from web crawlers (which can sometimes inadvertently highlight sensitive administrative directories).

The sitemap.xml provides a comprehensive map of all public-facing URLs, giving an attacker a complete layout of the site's intended structure without needing to brute-force directory discovery.

Part G: Reconnaissance Report (Conclusion Draft)

During the passive reconnaissance phase of this assessment, I learned that a significant amount of critical infrastructure data can be gathered without ever directly interacting with or attacking a target's core systems. By utilizing public WHOIS lookups and DNS enumeration, I was able to map out the domain's registration details, hosting providers, and mail server

routing. Furthermore, analyzing the HTTP security headers highlighted the vital role that server configuration plays in defending against common web vulnerabilities like Clickjacking and Cross-Site Scripting (XSS).

Reviewing the robots.txt and sitemap.xml files demonstrated how easily site structures and potentially restricted directories can be inadvertently mapped out by an external observer. Ultimately, this exercise reinforced the principle that successful ethical hacking relies heavily on thorough information gathering. By understanding the target's technological footprint—from its Next.js and Supabase architecture down to its DNS records—a security professional can identify potential misconfigurations and build a highly effective, targeted approach for subsequent security assessments